MONEY IN MESOPOTAMIA

BY

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Abstract

Although contemporary preconceptions about what money is or is not sometimes evoke doubt about the existence of money in ancient Mesopotamia, it seems clear that it did exist. Money was substance oriented, and coins, when they finally appear are weighed like any other valuable metal. The most common money substances were barley as cheap money and silver as the more expensive, but other substances were also used. As to the forms or shapes in which money circulated, a number of words in the ancient languages can be identified that probably refer to these forms, but their specific appearance remains, in most cases, unknown.

Was there such a thing as "money" in Mesopotamia? One of my younger friends in Munich—a superb Assyriologist who has read lots of Mesopotamian texts of various kinds—asked me this question in the summer of 1995 when I told him I was about to give a lecture on this topic in Wien.[1])

The question is a natural one. It is also a natural answer to another question: why is history always being rewritten? Not merely to incorporate "new evidence" but even more to explain the past to our contemporaries in words that they can comprehend. Most of us today think of "money" primarily as "paper money." From the fifth century B.C. through the beginning of this century, *coins* would probably have been the most common answer one would have gotten, if one could have made a random sampling of public opinion—at least, among Europeans and their colonists—on the question: "What is money?" However, as "our" millennium draws to a close, one can well imagine, not so far in the future, a world in which people will ask: "do you mean that those things were ever used as *money*?"

It is also a natural question, because, even though most of us think we can recognize money when we see it, defining what is money and distinguishing that from what is not money is a rather complex problem. Aside from the difficulties inherent in the subject itself, money—like religion, politics, morality, and certain other things—lends itself to dispute. This was summed up well

1) I record here my gratitude to colleagues in Wien for the opportunity of presenting a preliminary version of this paper as a lecture at the Institut für Orientalistik, Universität Wien, on 20 June 1995, and for their cordial and helpful comments.

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almost a century ago, in the twilight of a previous era in monetary history, by a well-known authority on the subject: "The difficult question as to the best definition of money has been complicated by the efforts of writers so to define the term as to give support to their particular theories."[2]) We, perhaps, who step with trepidation in the ever changing river of time can take some comfort, small it's true: some rivers remain—at least till now—much the same.

I do not discuss theory in this essay, though, of course, I recognize that all fact is theory. Moreover, I regretably have to say at the outset that, although it would delight me to be able to present a systematic overview of the history of money in ancient Mesopotamia, I cannot do so. There are no histories of money in Mesopotamia and very few reliable studies on which such an overview could be constructed. This is not because there is no evidence for money. Quite the contrary, there is a lot of evidence, both textual and archaeological, but putting this evidence together is not easy. I have had to content myself with a brief summary of the reasonably well attested functions of money and with a bit more detail about what we know concerning money substances and money forms.

Money, of course, did exist in ancient Mesopotamia. To answer in any other way would violate the basic principle of language, namely, communication, which prohibits the definition of words in defiance of common usage. Usage of terms like "money," "currency," "cash," etc. by cuneiformists[3]) to describe *silver* is so ubiquitous in the literature of the last century and a half that, if money were not recorded in cuneiform documents, one would have to make the improbable inference that everyone who had used this term had entirely misunderstood the texts. The non-specialist reader will find a sample of the literature relating to "prices"[4]) the best introduction to "money" but should be aware that the many variables involved make conclusions tentative and that the literature contains egregious blunders and perversities, as well as having its fair share of honest errors.

We turn first to *coinage*, because this brings us face to face with the problems and potentials of the Mesopotamian evidence, which reveals a complex economic system that functioned for centuries on end entirely without coinage. Coins turn up in the eastern Mediterranean in early sixth-century archaeological context[5]) and gradually begin circulating widely but are not archaeologically attested

2) Bastable 1911: 694.

3) E.g., Curtis & Hallo 1959; Ebeling 1964; CAD 1965 be'ulātu: 216f.; 1971 kaspu: 245-247.

4) Dandamaev 1988; Dubberstein 1939; Farber 1978; Gelb et al. 1991: 251-298; Kupper 1982; Lambert 1963; Limet 1972; Meissner 1936; Powell 1990; Renger 1989; Schwenzner 1915; Snell 1982; Steinkeller 1989: 121-138; Sweet 1958; Veenhof 1988; Zaccagnini 1988.

5) Calmeyer & Naster 1995.

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in Mesopotamia until well over two centuries later, at the end of the Achaemenid period.[6]) This apparent disinterest in coinage cannot be because Mesopotamia was an economic backwater. On the contrary, by the third millennium B.C., at the latest, Mesopotamia was crisscrossed by trade routes connecting it with the Asian land mass, the Mediterranean, the Indian subcontinent,[7]) and probably with east Africa.

Daniel Snell in a recent essay has remarked on this disinterest in coinage and urged that its use was not readily embraced by Near Eastern economies because it really offered nothing particularly new.[8]) Archaeological evidence confirming this judgment is provided by the coin horde discovered in the last century at Babylon by H. Rassam, which contained not only Greek, Phoenician, and Persian coins, but also broken silver[9])—obviously intended to be put in the balance pan along with coins in making money transactions.

That metal monies were weighed is what we expect from more than two millennia of cuneiform documentation. And, even in the Macedonian period, cuneiform documents contain records of money transactions in minas and shekels with the additional remark that the silver consists of staters of Antiochus or of other Macedonian rulers. These documents also evince a typically first millennium tendency to identify the quality of money with increasing precision by the addition of such terms as "burned [i.e., purified]," "worked," "good."[10])

These accounting remarks about silver coins—with their legal overtones—in cuneiform texts of the Macedonian period indicate two things: a concern for quality and routine weighing rather than counting. Weight—or size, in the case of grain—and quality were prime concerns in Mesopotamian money transactions from time immemorial. We return to the question of quality below in discussing money substances, but let us say a few words here about weighing.

Minas and shekels themselves were monetary terms in ancient Mesopotamia, as is evidenced not only by internal Mesopotamian usage but also by the fact that they turn up as words for monetary units in other languages. Originally, however, they were weight-metrological terms.[11]) The history of money in Mesopotamia is intimately bound up with weighing and measuring, and the bond between weighing and money persevered in the Near East until relatively recent times. We may recall W. Eilers' story about how, when he cashed a check for a large sum of money in 1933 in Bushire at the Imperial Bank of

6) Moorey 1994: 237; Calmeyer & Naster 1995.
7) Astour 1995; Potts 1995; Yoffee 1995.
8) Snell 1995: 1496.
9) Reade 1986; Calmeyer & Naster 1995.
10) CAD 1960 istatirru: 204; CAD 1982 qalû: 68f.; CAD 1958 epšu: 246f.; CAD 1965 babbanû: 7f.; Powell 1978: 217.
11) Powell 1990a: 508-517.

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Persia, it was weighed out to him on an "uralten grossen Hängewaage" and that, on transporting it to the Persian National Bank, it was found to contain precisely the intended amount of money.[12]) Weighing of coined money—particularly high-value money like silver or gold—has probably always been more ubiquitous than generally thought, and we may note in this context that one of the firm points of agreement in the interesting debate between M.L. Bates and P. Balog about the function of Fāṭimid "glass weights" was that coins were, in principle, weighed rather than counted.[13])

Monies in ancient Mesopotamia were themselves all potentially useful substances. In other words, except for the possible use of "axes" (noted below near the end of this essay), we have no evidence for symbolic or representational monies, and this substance-oriented character of the monetary system probably played a significant role in the remarkable long-term stability of prices in ancient Mesopotamia.[14]) However, money is part of a total social package. Mesopotamian society seems to have been fundamentally hierarchical and was probably essentially hostile to innovation, an attitude with which I find myself increasingly sympathetic (and Near Eastern resistance to the European paradigm continues but is probably doomed). Moreover, the majority of the ancient Mesopotamian population was probably rather poor. These are the people we never really see in cuneiform documents. They occasionally appear as numbers, far less often as names, but their labor must have played a major role in the whole economic system and cheap labor must have played a greater role in price stability than the kind of money that was used.

Mesopotamian monies are distinguished from ordinary commodities by being exchanged, sometimes for one another, more often as payment for other tangible commodities or for other less tangible goods, such as freedom from obligations of various sorts (taxes, loans with interest, fines, and so on), and at more or less stable rates. They were used as indices or standards of value and were stockpiled as storable wealth. Some of them, like barley and silver are well attested as standards of value and monies of account in documents of nearly all periods and areas. Barley, along with silver, plays a prominent role in mathematical problems about money in Old Babylonian mathematical texts,[15]) which reflect the concerns of the era from about 2200 to 1700 B.C.

Materials that clearly functioned as money are, to arrange them in ascending order of value: barley, lead, copper or bronze, tin, silver, gold. Of these barley, lead, and copper or bronze functioned as cheaper monies; tin was mid-range;

12) Eilers 1957: 328.
13) Bates 1981, esp. 68ff.; Balog 1981, esp. 101f.; cf. Bivar 1982.
14) Powell 1990.
15) Nemet-Nejat 1993: 183f., 225f.

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silver and the much rarer gold were high-range monies. This is not to say that other physical objects or substances did not occasionally function as money: cows, sheep, asses, slaves, household utensils, and everything under the sun occasionally had a money function. The notable thing about "normal" monies is that these—above all silver and barley—are over and over again used as common denominators for value.

Mesopotamian money transactions from "prehistoric" times onward were not "barter economy." To insist that "money" can only be "money'' and nothing else is merely another example of defining everything to fit preconceived theories, a favorite pastime of those who, supposedly, use their brains rather than their hands to perform "work." The Mesopotamian evidence vitiates scholastic definitions like this. Barter must have taken place everywhere and often in ancient Mesopotamia and is probably the major reason we have so little evidence for exchange of cheap commodities—especially in small amounts—in cuneiform documents. Daniel Snell has made the excellent point that "barter" and "marketless trading" cut across time and are still very much alive.[16]) That will not surprise the dwindling few who are familiar with rural life at first hand. My mother, who spent her whole life in western Tennessee (USA) and completed her twelfth septad on 24 September 1995, was an expert at "primitive" economy and "traded" almost every conceivable kind of thing, from tangibles to intangibles, without written documentation and usually without bargaining. This was, moreover, conducted in tandem with other "market" trading, sometimes with the formalities of documentation. The historian could never retrieve this contemporary "primitive" economy from the documentary record. It emerges only—and that rather erratically and intractably from—narrative sources, exactly what we lack for ancient Mesopotamia. On the contrary, it is precisely the "monetary" economy that we do see in the documentary record. That is the reason we have it: it concerned "weighty" matters like minas and shekels.

We should note, of course, that these Mesopotamian monies do not form a single "system." Preferences varied according to time and place. Silver and grain probably have to be regarded as "basic" monies and would probably have been acceptable everywhere, within certain bounds.

Grain was more subject to value fluctuations, caused by season, by weather, or by any of the other forces that affected agriculture in ancient times, and this is surely why interest on barley loans was usually one-third of the principal, whereas silver was only one-fifth. Grain probably would not have been acceptable, as a rule, for making large purchases or paying large debts. It should be

16) Snell 1995: 1487f.; Snell 1991.

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noted, however, that the temples in Babylonia from the Ur III period through the Achaemenid period use barley like money, especially as a unit of account. These institutions were in a position to store grain. They needed it to feed their dependents, and it is clear that they could turn it rather easily into all sorts of other things they might need, ranging from labor-services to commodities. Another limitation on grain is that it (like everything else both cheap and bulky or heavy) was not viable as a money (or trade item) in long-distance overland trade—academic theorists sometimes overlook this salient point. Barley functioned as a cheap, local money, especially where the other primary money was silver. Ancient technology did not permit sufficiently accurate weighing of small amounts of silver and, moreover, placed the payer of silver at the mercy of the receiver unless both were equipped with weights and balances. Silver in amounts less than one-shekel (about 8.33 grams) probably rarely changed hands, for the simple reason that even this small amount of silver constituted about a month's pay for labor throughout a large part of ancient Mesopotamian history.[17])

In Babylonia, silver and barley served as basic monies from at least the mid-third millennium on. Copper or bronze (probably both: distinguishing between these remains problematic) is attested for a time in the third millennium (in the so-called Fara and Presargonic periods) as a low to mid-range money, then practically disappears, turns up again in the Kassite period, but was perhaps used more widely as money than our present evidence indicates. Gold, which in general is rare, seems to have functioned rather commonly as money in the later Kassite period (14th-13th centuries B.C.), but the reasons for this as well as the precise character of this "gold" remain obscure.[18])

In the north (Assyria), there seems to have been a predilection for cheap metal monies, perhaps reflecting the greater dependency of the north on rainfall agriculture, which in turn would have caused far greater fluctuations in the value of grain than in the irrigation agriculture of Babylonia. In the north, lead often turns up as the "cheap" money rather than grain, but copper/bronze also occurs and tin plays a mid-range role that is never found in Babylonia. Silver is usually the high-range money, and gold, which appears sporadically here and there in a money function, also turns up in joint-ventures from the early second millennium B.C. as an extra-high-range money in ways that are alien to the Babylonian pattern. On the other hand, the material from Mari on the Middle Euphrates in the 18th century B.C., which reflects a palace economy, looks

17) Powell 1979: 83-87, 1990: 96-98.
18) Powell 1990; Brinkman 1987, 1988; Moorey 1994: 219.

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much more like the Babylonian paradigm, where silver is the standard metal money.[19])

One peculiarity about the documentary evidence for Mesopotamian monies should be noted: the general absence of specifications about *quality*. This ambiguity, however, is characteristic of cuneiform documents as a whole, and, although there is an observable tendency toward more precision over the roughly two millennia from ca. 2400 to 400 B.C., into which falls most of the surviving and intelligible documentation, we are frequently reduced to guessing about why one piece of real estate is worth more than another or why one slave or animal is worth only half of another. It is one of the major problems in trying to study trends and prices in Mesopotamian economic history.[20])

The underlying—but rarely stated—assumption about prices or stated values of commodities is that "money," if not qualified in some way, means a "standard" kind of money, presumably "pure." With silver and barley, this assumption is probably justified in most cases, but proving it will probably always be impossible, because there is no way to avoid circular arguments.

Still, the problem of the quality of grain and silver is simple compared to that of copper and bronze or lead and tin or even gold. The terminology for copper and bronze is so ambiguous in Sumerian and Akkadian that we are still far from solving this problem, and the problem is made especially intractable by the nearness of copper and bronze on the price scale.[21]) Moreover, although it seems certain that the Akkadian word *annaku* (*anāku*) and the "Sumerogram" an.na, with which it is often written, usually denote "tin," it also seems equally certain that this is not always the case. The ancient Mesopotamians did not, of course, confuse lead and tin. Quite the contrary: "pure" tin had a value about forty times that of lead. However, occasionally cheap metals similar in appearance to tin seem to have been called by the term for tin.[22])

Gold, which occurs as a money of account for large sums in Assyrian business ventures from the early second millennium,[23]) is—because of its high value (normal range being about 6 to 15 times that of silver)—usually a commodity rather than a money. Like silver, it is not regularly described in any way that gives us a clear idea about its quality. When it does function as a money in the late Kassite period (14th through early 12th centuries), gold that is "red" (if that is really what the term means) is twice as expensive as gold that is "bright,"

19) Powell 1990; Müller 1982; Veenhof 1988.
20) Powell 1990; Gelb et al. 1991: 251-298; Steinkeller 1989: 121-138; Dandamaev 1988; Snell 1982; Farber 1978.
21) Brinkman 1988: 136-139; Powell 1990: 79-87; Moorey 1994: 242, 254.
22) Freydank 1982; Müller 1982; Powell 1990; Moorey 1994: 295f., 298f.
23) CAD 1980a naruqqu: 381f.

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exactly opposite of what we would expect from our color perspective.[24])

We cannot presently solve the quality problem. Perhaps it is ultimately insoluble. However, the nature of the problem and a good sample of what we probably can and cannot learn about money can be illustrated—I hope, in a way that non-specialists can understand—by examining a common expression in Chaldean and Achaemenid period texts: *kaspu ša ina ištēn šiqli bitqa*, "silver of which (there is a) *bitqa* in one shekel." It is now generally agreed that this expression means: silver which contains an alloy of one-eighth,[25]) and there are parallel expressions *kaspu (ša) ina ištēn šiqli šitta girê*[26])/*ḫummušu*[27]), "silver (of which there are/is) two carats [= two 24ths]/(a) fifth in one shekel." Alloying, however, is inferred from context. Neither a word for alloy or alloying nor the metal used for alloying (usually assumed to be copper on the basis of modern analysis of ancient silver objects[28]) has yet been identified in connection with monies.

These expressions are by no means easy to interpret, as is shown by the fact that the Chicago *Assyrian Dictionary* [CAD]—after years of interpreting them as above—in its treatment of the word *nuḫḫutu*, reverted to an old reading *kaspu ša 1/2 šiqil pitqa nuḫḫutu* and translated "silver of *n.* quality, cast in half-shekel pieces."[29]) This phrase, however, has to be read *kaspu ša ina ištēn šiqli bitqa nuḫḫutu*, "silver of which there is a *bitqa* in one shekel *nuḫḫutu*-ed." But why?

First, the two cuneiform wedges that the *Assyrian Dictionary* has read as "1/2" are, in reality, a horizontal plus a vertical and would be seen to be distinct from "1/2" if one examined the signs on the tablets (the hand copies by which many, but not all[30]) Assyriologists, swear—in vain—constantly muddle fine distinctions like this and are the bane of a metrologist's life). Second, despite Sidney Smith's argument[31])—oft repeated (also by me)[32])—that casting of "half-shekel pieces" is mentioned in the inscriptions of Sennacherib, nothing is actually said about casting half-shekel *pieces*, and the phrase in Sennacherib's inscriptions[33]) is probably just hyperbole implying that the huge bronze figures were cast *as though* they weighed only half a shekel each. In short, half-shekel

24) Waetzoldt 1985; Powell 1990: 80-82; Moorey 1994: 217f.
25) CAD 1965 bitqu: 278.
26) CAD 1956b girû: 96f.
27) CAD 1956a ḫummušu: 235.
28) Hughes 1986: 88; Moorey 1994: 240.
29) CAD 1980b nuḫḫutu: 318.
30) Civil 1994: 22 n. 11.
31) Smith 1922.
32) CAD 1961 ze'pu: 87; Powell 1978: 226.
33) Luckenbill 1924: 109:18, 123:29.

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pieces have never been clearly demonstrated in cuneiform records from any era. Third, though the reading *pitqa* (with putative derivation from *patāqu*, "to form or cast") is theoretically possible, it is improbable, because *bitqa* (one-eighth) is paralleled in the similar expressions *šitta girê* (two twenty-fourths) and *ḫummušu* (one fifth), and all three are metrological-monetary terms.[34])

Turning now to the term *nuḫḫutu*, we may begin by noting that it has generally been interpreted as denoting a quality of silver.[35]) However, as I pointed out some years ago in discussing the etymology and usage of the term, the expression "of which there is a *bitqa* [one eighth] in one shekel" already defines the *quality* of the silver. I also called attention to the "cutting" implicit in the long-postulated Semitic cognates of *nuḫḫutu*—reaffirmed in a recent discussion of the word[36])—and to the semantic parallel in Greek *koptō*, which is used in a technical sense of coining money, and I pointed out that absence from the archaeological record does not mean that Mesopotamia was ignorant of coinage in the Chaldean and Achaemenid periods.[37]) I am now, however, inclined to believe that *nuḫḫutu* is the technical term for "alloyed" in a monetary sense. It cannot itself be a quality term but must have some technical meaning associated with turning silver into money-silver. The reasons for this conclusion will become apparent shortly.

The qualifications specifying that silver contains, per shekel, *ḫummušu* (1/5), *bitqa* (1/8), and *šitta girê* (2/24) correspond to fineness grades of .800, .875, and .916. What is most interesting about these fineness grades is neither the fact that they existed nor the relative fineness—interesting though these may be—but rather the minuteness of the distinctions implied. Several interesting conclusions may be drawn from this.

First, whereas a silver of .800 fineness might be distinguishable by eye from a "pure" silver (depending upon what the alloy was, and I must repeat that we cannot be sure of that), it is wholly improbable that the eye alone could distinguish between these various grades of fineness.[38]) This means that the fineness of the silver must have been marked in some way.

Second, the Babylonian phrases "fifth in a shekel," "*bitqa* in a shekel," and "two carats in a shekel" are *fineness* terms. Ancient balances as a whole were not capable of measuring accurately these small masses (postulated values: ca. 1.66, 1.14, 0.69 g.). Thus, the whole fineness terminology implies that someone somewhere was alloying silver by mixing one mina (ca. 500 g.) of alloy with

34) Powell 1990a: 512f.
35) Soden 1958-1981: 810; CAD 1980b nuḫḫutu: 318.
36) Tropper 1995.
37) Powell 1978: 223-226; cf. Reade 1986: 85.
38) Reade 1986: 86.

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four of silver to create the .800 fine silver, one mina of alloy to seven of silver to create the .875 fine silver, and one mina of alloy to eleven of silver to create the .916 fine silver, and, moreover, marking the resulting alloys in some way for fineness.

The most commonly occurring designation for fineness is *bitqa* (one-eighth), and this is very frequently followed by the term *nuḫḫutu*, for which I have suggested—tentatively—the meaning "alloyed" in the technical sense of silver that has been alloyed and graded to a specified fineness. One should, therefore, assume that, in Chaldean and Achaemenid texts, the term *nuḫḫutu* is implicit wherever money silver other than "pure" occurs and that its omission is mere ellipsis. To the expression *kaspu ša ina ištēn šiqli bitqa nuḫḫutu*, is occasionally added another term, *ginnu*,[39]) which occurs in both positive and negative forms, i.e., "*nuḫḫutu*-ed silver of which there is a *bitqa* in one shekel, which is/is not *ginnu*." This interesting term also gives us a further glimpse into the enigmas and possibilities involved in trying to write the history of money from cuneiform documents.

It has generally been assumed that the expression "which is *ginnu*" means that the silver was stamped with a mark denoting its *quality* and that the negative expression "which is not *ginnu*" means that the silver in question was not marked for *quality*. This, however, cannot be right, because, the quality of the silver is already specified in the designation of fineness. And, since a fineness of seven parts pure silver to one part alloy could not possibly have been determined merely by looking at an amorphous mass of alloyed silver, the fineness must have been indicated in some way, either by a stamp or by the form (shape) in which the silver circulated or by some other visible means.

Consequently, since *ginnu* cannot denote quality, it probably refers to the other obvious thing one looks for in such cases: the mark of guarantee. The orthography of the word, occasionally written gi.na, makes it likely that the ultimate origin of the word is Akkadian *kīnu*, having a meaning of "true" and related senses; the Akkadian word seems to have been borrowed into Sumerian before the end of the third millennium then borrowed back into Akkadian several times in specialized senses, which creates all sorts of problems for lexicographers.[40]) Connection of *ginnu* with *kīnu* seems especially probable, because gi.na/kīnu occurs on weight stones from the Ur III period (21st century B.C.) onward in the sense of "made true," usually with the name of the guaranteeing authority.[41])

But what is being guaranteed about the silver? Certainly not the weight but

39) CAD 1956b ginnu: 79f.; Soden 1958-1981: 290.
40) CAD 1956b ginû: 80-82; 1971 kīnu: 389-393; Reade 1986: 85.
41) Powell 1971: 249-254, 256-263.

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rather the quality. However, the *ginnu*-guarantee cannot have been the *mark* of quality itself, because, as we have already observed, the *quality* of silver described as being "not *ginnu*" is specifically stated to be known, i.e., .875 fine. This can hardly mean anything else except that silver circulated in Chaldean and Achaemenid Babylonia in customary forms or with marks that would have enabled someone who customarily dealt in silver to recognize its fineness and that in many cases this was further marked in some way with the "seal" of some authority.

That all of this must remain so speculative describes the problem of cuneiform records in a nutshell: nearly everything that was obvious to them was not written down. Mesopotamia in the time of Darius (522-486 B.C.) was no less the center of a vast empire than at the height of the Abbasid caliphate. It has been generally agreed upon for years that the Daric was footed upon the Babylonian shekel,[42]) but we still have no solid evidence for why, and, moreover, Greeks who wound up in Mesopotamia, like the soldiers who accompanied Cyrus the Younger in 401 B.C., must have paid with coins for the purchases of food and drink described by Xenophon in the *Anabasis* (1.5.10, 2.4.28), yet the first recognizable word for "coin" in cuneiform is *istatirru*, a loanword from Greek *statēr* in the Macedonian period.

To reiterate: the quality of money substances is usually passed over unmentioned throughout most of the cuneiform record. This does not mean that ancient Mesopotamians were unconcerned or unaware of quality but precisely the opposite: they knew exactly what was going on and did not feel it necessary to record the details. The increasing precision with which the quality of silver is noted in the Chaldean-Persian period may indeed reflect a heightened concern for fineness in silver, but it cannot be legitimately interpreted as something wholly new. Moreover, the .875 fineness grade is attested before the end of the reign of Nabu-kudurri-uṣur[43]) ("Nebuchadnezzar") II (605-562 B.C.) and suggests an awareness of the money value of silver and of the monetary significance of alloying that is without parallel in the numismatic or written evidence for Greece in this era. Precisely what this fineness phenomenon means remains to be clarified.

One final point about difficulties. Evidence about "money" tends to cut across text genres, and the content of this evidence is technical and hard to interpret, even for those who have spent years working at it. In studying money, prices, and analogous things, the single most labor-costly element is error, not ancient error, which is often relatively easy to pinpoint if one is looking at the tablet

42) Powell 1971: 193, 250-252; 1990: 510f.
43) E.g., Dandamaev 1984: 525 + 665f.

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or a good photograph, but the errors of modern copyists. Miscopied numbers are the single most time consuming aspect of studying Mesopotamian metrology and money, but it is not limited to numbers. The copy of a cuneiform text dated in the 94th year of the Seleucid era published over a century ago by J.N. Strassmaier begins: "two-thirds mina of silver, purified [*qalû*], *šul*?-*šú*, good [*babannû*], staters of Antiochus."[44]) However, it is clear that what the tablet must have is: "two-thirds mina of silver, purified, worked [*ep*^{1}-*šú*], good, staters of Antiochus." My point is this: even with parallels for the correct reading[45]) and with Strassmaier's faint indication that something is wrong with the sign copied "šul" (probably an ancient correction), both Akkadian dictionaries nevertheless interpreted the word read "*šul*?-*šú*" as "one-third"—even though it makes no sense whatever.[46]) We are still struggling with too many problems like this, which is why writing a reliable history of Mesopotamian money remains a task for the future.

The forms or physical shapes that money took are even less well understood than the qualities of the material substances that were used as money. This is primarily because of the laconic nature of cuneiform documents and the lack of archaeological evidence.[47])

Archaeological evidence is scarce for several reasons. One is that metal monies were never thrown away to be discovered in a trash heap. Even the relatively cheap ones would normally have been recycled. So we cannot expect an abundance of this type of artifact. A second (frequently overlooked) is that Mesopotamia simply has not been as extensively or intensively explored as many other areas of the Mediterranean. A third is: how are we to recognize "money" when we see it? Coins are relatively easy to recognize, and they are duly registered as such in excavation reports and in museum catalogues, but the basic paradigms for Mesopotamian monies still remain to be established.

It seems likely that more Mesopotamian money has been discovered and has made its way into museum collections than is currently known. However, the only labor efficient way to study this material is by systematically going through all of the metal finds in museums with Mesopotamian artifacts and comparing that information with the cuneiform evidence. This, however, would be a gigantic undertaking, even in an ideal world, and is not likely to be done any time soon.

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44) Strassmaier 1988: 137, 150.
45) CAD 1982 qalû: 68.
46) Soden 1958-1981: 1269; CAD 1992 šulšu: 258f.
47) Moorey 1994: 238.

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The one form about which there seems now to be some agreement is the so-called "ring" silver.[48]) "Ring" is a translation of the Sumerian word *ḫar*, corresponding to Akkadian *semeru*. Both words may have also meant ring in the sense of a continuous circular band; however, from the Ur III period through Old Babylonian (ca. 2100-1600 B.C.), these terms seem to have denoted primarily bands of silver shaped into coils or spirals.

I chanced upon this "ring money" while weighing weight specimens at the Oriental Institute of the University of Chicago about two decades ago, because Judith Franke, a Mesopotamian archaeologist, called my attention to them, and I was able to recognize their significance because I happened to know about the Ur III and Old Babylonian "rings."[49]) Others had surmised previously that "ring money" ought probably to mean something like this,[50]) but no one had ever put it together because much of the archaeological evidence I used does not come from regular excavations, and, to judge from published works, it has not been common in Mesopotamian finds. I have since observed wire coils displayed in museums from the Mediterranean to Scandinavia, some of which evinced the same kind of "beaten off" character that I had observed in the Mesopotamian silver coils,[51]) probably the result of hacking off a piece to "pay" for something.

This practice of cutting coils naturally brings to mind the "Ring Breakers" of ancient Germanic literature.[52]) Bronze coils very similar in type have been found widely in Europe and occur ubiquitously in the volumes of *Prähistorische Bronzefunde*. The historical implications of this are not clear to me, but perhaps there is ultimately some kinship link with the ancient Near Eastern money coils, which probably go back at least into the fourth millennium B.C. Finally, we should mention that the symbol[53]) in the earliest cuneiform script (ca. 3000 B.C.) which is the ancestor of the symbol that stands for the word *kug*, "precious metal," in Sumerian, may represent a stylized picture of a metal coil or spiral.

Another form of money that has been tentatively identified in cuneiform records are "cakes" or "loaves." Lipiński[54]) has argued that "lead-cakes" discovered at Ashur represented forms of money, proposing to identify them with the Akkadian word *akalu*, "bread(-loaf)," and has noted as a likely parallel the silver "cakes" discovered at Sam'al (modern Zincirli in the ancient Cilician area of

48) Snell 1995: 1488f.; Moorey 1994: 237; CAD 1984 semeru: 223-225; Michalowski 1978. Cf. Einzig 1966: 190f.
49) Powell 1978: 211-222.
50) Powell 1978: 234.
51) Powell 1978: 241 pl. II c. Cf. Bivar 1971: 102f. + pl. IV; Einzig 1966: 191.
52) Eilers 1956: 336; Powell 1978: 234.
53) Green et al. 1987: 233 no. 301.
54) Lipiński 1979: 574f.; cf. Bivar 1971: 100.

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modern Turkey). However, whether the lead objects found at Ashur were really the "cakes" of the cuneiform records remains uncertain, and Roger Moorey has recently stated that lead ingots are still unknown from Mesopotamia.[55])

Identifying other forms of money in cuneiform documents is still more problematic. Akkadian words that may denote some form of "ingot" are: *ganūnu*,[56]) *gubāru/kubāru*,[57]) *kušru*,[58]) *lišānu*,[59]) *ṣīdu*,[60]) and *šibirtu*.[61]) However, the specific form or forms to which these words refer cannot be identified with certainty. One could speculate that *lišānu*, "tongue," is the name for the tongue-shaped little ingot of silver found together with other silver objects in a jar at Ešnunna in Akkad period context; however, there is no assurance of this: "tongue" also describes the blade of a knife or sword and the share of a plow.[62]) Moreover, with which of these words we should identify other archaeologically recovered "ingots"[63]) remains wholly uncertain.

The term *šibirtu*, which ought to mean, according to both its own Semitic etymology and its Sumerian equivalent, "broken,"[64]) is a good example of how difficult it is to identify cuneiform money terms with archaeological artifacts. The evidence assembled by the Akkadian dictionaries shows that this term eventually came to denote something on the order of "ingot," because texts from the mid-first millennium speak of *šibirtu* silver as being precisely one mina each and some describe the silver as being "*nuḫḫutu*-ed with a *bitqa* per shekel," i.e., .875 fine. Clearly *šibirtu* cannot mean in such cases "broken silver." But perhaps it did start out as such and, from the practice of melting down broken silver into more regular forms, then evolved into the concept of an ingot that had been formed from "broken" silver. If not, what did one call in Sumerian or Akkadian the "broken" silver that turns up in archaeological finds?[65]) Possibly *ḫušû*,[66]) a word that seems to denote metal scraps, but, unfortunately it is not regularly associated with silver. The Oriental Institute Museum, University of Chicago, possesses silver of this type (as well as "rings") from the Diyala region—some purchased, some excavated, and still unpublished—which,

55) Moorey 1994: 293.
56) CAD 1956b: 43 + 1984 sikkuru: 256.
57) Soden 1958-1981: 295; CAD 1971: 481f.
58) CAD 1971: 600; Brinkman 1988: 138f., 152 n. 19., 155 n. 46.
59) CAD 1973: 215.
60) CAD 1962: 174; Soden 1958-1981: 1100.
61) CAD 1992: 379f.; 1984 siparru: 297.
62) Powell 1978: 242 pl. III B; Oriental Institute nos. A8901 (pot), A8767-8772 (contents). Cf. Einzig 1966: 191f.
63) E.g., Moorey et al. 1988: 46f.
64) Powell 1978: 222f.; Eilers 1956: 337; cf. Snell 1995: 1489.
65) E.g., Powell 1978: 242 pl. III A; Arnaud et al. 1979: 15.
66) CAD 1956a: 262.

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according to Museum records, goes back into the Early Dynastic period, some even back as far as Early Dynastic II (putative date ca. 2700 B.C.), but we do not know what it was called.

Two other forms in which money circulated were cups[67]) and jewelry,[68]) and occasionally these two are contrasted with plain "silver,"[69]) obviously referring to one of the usual forms in which silver was traded, but the form is not specified. Perhaps the smaller coils made of wires that are round in cross section are hidden under *sāḫertum*[70]) or *ṣibarātum*,[71]) despite the meanings assumed for these words by the dictionaries, and some form of money—but what is quite unclear—must be represented by *samrūtu*,[72]) tentatively interpreted as "nail" on the basis of etymology.

One suspects, of course, that one or more of these money-form words describes at least some of the silver objects recently discovered in the Akkad period level of Tell Brak[73]) or with the "ingots" from a hoard in western Iran dating to around 600 B.C.[74]) But which?

Before concluding this inconclusive discussion of money forms, let us note one remarkable oddity: the Sumerian sign for "shekel" actually seems to be the stylized picture of an axe; moreover, the Sumerian word *gin* not only means shekel[75]) but also "axe."[76]) This can hardly be accidental. Axe hordes are ubiquitous, and it is virtually certain that these were used as money,[77]) and, indeed, axes are on rare occasions attested in cuneiform documents as a money form.[78]) Mesopotamians may have begun—at least in part—where we have ended, in representational money.

Perhaps that sums up best what is most striking about the Mesopotamian evidence concerning money. It defies classification under simple rubrics. Deposit, credit, joint ventures, orders to pay and many other aspects of "modern" monetary economy can be glimpsed here and there rubbing elbows with "primitive" or "archaic" practices. Above all, the Mesopotamian evidence challenges the idea of some kind of ineluctable, inevitable evolution toward more "rational" forms of money as well as the often alleged importance of coinage for the

67) CAD 1971 kāsu: 253-256.
68) CAD 1959 dudittu: 168-170.
69) CAD 1971 kaspu: 245.
70) Soden 1958-1981: 1008; CAD 1984 saḫirtu: 57-59.
71) CAD 1962 ṣibārātu: 154.
72) CAD 1984: 120f.
73) Oates & Oates 1993: 162-168, 185-188.
74) Bivar 1971: 101f. + pl. I; 1982; Curtis 1984: 1-22 + pl. V-X.
75) CAD 1992 šiqlu: 96.
76) Soden 1958-1981: 846 (pāšu).
77) Einzig 1966: 189f., 231; Moorey 1994: 237.
78) CAD 1956b ḫaṣṣinnu: 134.

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economy. They, obviously, were not impressed by it. Someday—though not soon—we may be able to say why.

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